

Blockchain Application: Healthcare



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1. The Current Situation (1/2)

Digital records and many stakeholders shape today's healthcare landscape

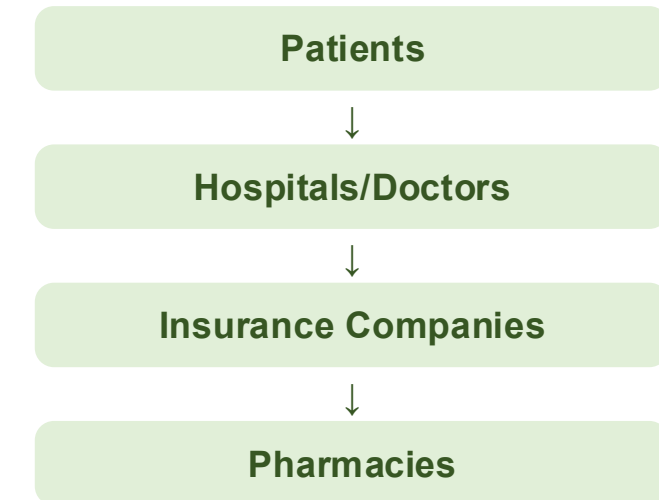


Digital Healthcare

- Usage: Electronic Health Records (EHRs), & Electronic Medical Records (EMRs).
- Record information: Diagnosis, Prescriptions, & Laboratory results.



MULTIPLE STAKEHOLDERS



Data exists, but is spread across different systems.
Question: **What current situation creates problems for healthcare?**

1. The Current Situation (2/2)

A Real-World Example of the Risks of Centralized Healthcare Data



Source: Reuters (2025); HHS (2025)

192.7
MILLION PEOPLE

\$2.2
BILLION



Exposed / stolen:

- Medical records & health information
- Insurance information
- Personal information
- Billing & payment information

2. Challenges

Fragmented data, weak data security and limited patient control



a) Fragmented data:

Different systems



Difficult data share

Example: The doctor cannot quickly access all of the patient's relevant medical history → More difficult decisions making → Inappropriate treatment.



b) Data security:

Centralized database



Cyberattack

Example 1: 2018 SingHealth cyberattack in Singapore made **1.5 million patients' information** compromised.



c) Patient control:

Who has accessed their health information?



When was it accessed?



How is it being used?



Limited visibility & control.

Three important goals: **sensitive data protection, effective exchange of information, & patient access to their data management.**

Question: **Can blockchain help address these challenges?**

3. Blockchain Technology as a Solution

Moving from a vulnerable, centralized model to a distributed, patient-owned one



Fragmented & Distributed

- Central hospital server = single point of failure
- Encrypted and distributed across a peer-to-peer network
- Standardized protocols let different hospital systems "speak the same language"



Patient-Centric Health Records

- Decoupling personal health records from individual hospitals
- Data is consolidated into a single, unified profile controlled directly by the patient

Outcome: Blockchain moves health data from institution-owned silos to patient-owned records

4. Deep Dive Key Use Cases (1/2)

Core applications for modern healthcare administration



Patient-Led Access

- Patient holds private/ public cryptographic key pair
- Medical data stored off-chain (hash on)
- Doctor requests access → smart contract → patient approves
- Every request permanently logged



Accelerated credentialing

- Institution digitally signs credential → written to blockchain
- Instant verification via public key
- Smart contracts automate the entire process



Remote Monitoring

- Wearables continuously capture readings
- Gateway timestamps only hash goes on-chain
- Smart contract checks readings against medical thresholds
- Automatic real-time alert to clinician

Sources: Haleem et al., 2021; Ray et al., 2021; Al-Sahan et al., 2020; Golinelli et al., 2022; Elangovan et al., 2022; Oracle, 2023; TMA Solutions, 2024; Roche Diagnostics, 2025;

4. Deep Dive Key Use Cases (2/2)

Nationwide Blockchain-Based Digital Health Ecosystem

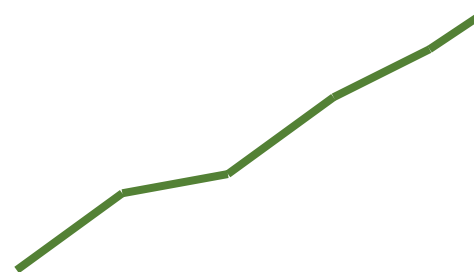
First nationwide digital healthcare system based on blockchain technology

- Integrating every citizen's EHR on a distributed network
- streamlined insurance claims
- no redundant diagnostic testing
- full patient data sovereignty and integrity
- cross-border care across 11+ EU countries

Estonia Records Secured

40M+

HEALTH RECORDS SECURED IN
ESTONIA



Outcome

- Patient-controlled databases, restore data ownership, accelerate emergency care and enable secure collaboration between institutions.
- Blueprint for other healthcare systems

Sources: Fatoum et al., 2021; e-Estonia, 2026

5. Blockchain System

Creating a Shared Source of Truth

Traditional system

 Hospital A → own database

 Hospital B → own database

 Laboratory → own database

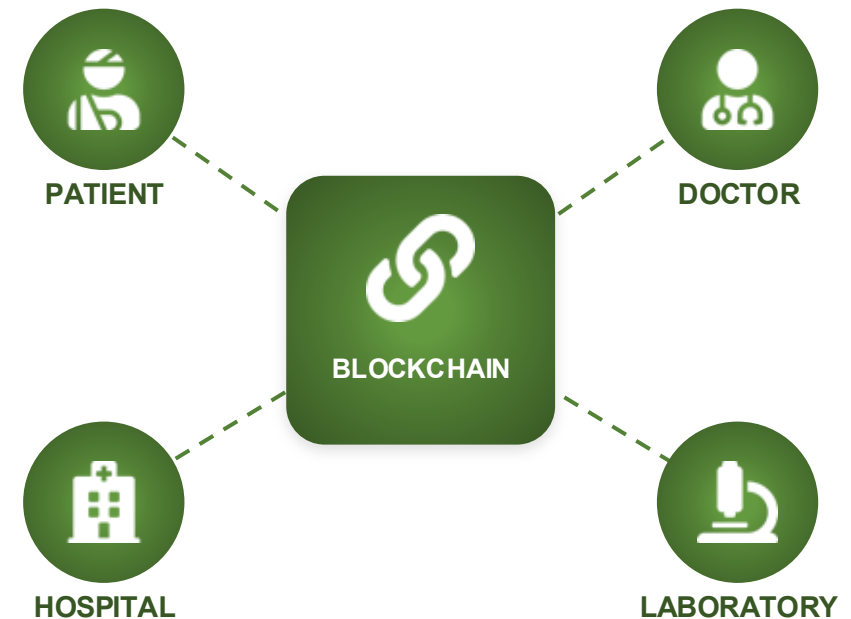
 Pharmacy → own database

Problems:

- Fragmented information
- Difficult data exchange
- Central points of failure
- Limited transparency

Blockchain-based-system

Decentralized | Transparent | Tamper-evident



Outcome: Blockchain enables secure and verifiable sharing of medical data without relying on one central database.

6. Smart Contracts

Enabling Patient-Controlled Access



Automatic: No manual approval process

Conditional: Access only under predefined rules

Traceable: Access events can be recorded

Outcome: Smart contracts automate who can access patient information, under which conditions and for how long.

7. Advantages vs. Disadvantages

Blockchain in Healthcare Benefits Come with Trade-offs

ADVANTAGES

✓ Better Data Sharing

✓ Data Integrity

✓ Patient Control

✓ Automation

✓ Traceability

DISADVANTAGES

! Privacy Concerns

✗ Difficult Data Correction

✗ High Implementation Effort

✗ Limited Scalability

✗ Smart Contract Risks

Outcome: Blockchain can improve trust, control and automation, but privacy, scalability and implementation remain major challenges.

Sources: Agbo et al. (2019); Kuo et al. (2020);

8. Moral Hazard (1/5)

Concealed actions and unverified data create room for misuse

Concealed Action

Insurers and regulators cannot physically be inside every exam room or research lab.

Information Asymmetry

The system relies entirely on self-reported, easily manipulated data



Stakeholders take advantage on blind trust and the system's lack of transparency

8.Moral Hazard (2/5)

Example:

Two typical cases of moral hazard in healthcare



Provider-Side Upcoding (overbilling)

Clinic bills insurance for an expensive procedure, but actually it's just a basic check-up.



Unverified Claims & Data

Pharma companies hide negative side effects from the clinical trial to a drug gets approved.

Sources: Arrow, 1963; Krugman & Wells, 2015, Silverman & Skinner, 2004, Haleem et al., 2021

8. Moral Hazard (3/5)

Smart contracts as automated auditors against upcoding

Smart contracts works as fraud eliminators.

Reducing upcoding: smart contract act as automated auditors by matching claims with immutable treatment ledger.

Example: A clinic bills \$500 for a remote heart-rate intervention

- Blockchain Action: Smart contract cross-references the claim against patient's wearable data which is immutable on ledger.
 - => abnormal heart rate is not recorded so contract instantly deny the claim (fraud prevented).
 - => if the device recorded abnormal data, the claim will be accepted.

8.Moral Hazard (4/5)

Verifiable research goals and end-to-end traceability against fraud



Eliminating scientific & supply chain fraud:

Before starting research, scientist must state their goals to prove on blockchain => preventing manipulation outcomes



End-to-end provenance:

traceability from factory to pharmacy shelf => avoid financial incentive to distribute fake drugs.

Sources: Haleem et al., 2021, Mackey et al., 2019, TMA Solutions, 2025

8. Moral Hazard (5/5)

Can Blockchain completely eliminate moral hazard?

Effect of Blockchain on Healthcare:

preventing fraud by shifting payoff matrix mathematically and transparent record

Can it eliminate moral hazard completely

cannot completely eliminate moral hazard which underlying human behavioral incentives

Example: A home-care nurse uses a blockchain-backed GPS app to log a 2-hour paid medical visit to an elderly patient.

- Nurse checks in but sits on their phone without actual care and then checks out.
- Result: blockchain flawlessly records a verified 2-hour visit and triggers payment.

=> Blockchain secure digital event but cannot police physical human behavior.

**Blockchain reduces moral hazard,
but cannot eliminate it completely**

9. Wrap Up:

PROBLEM

- healthcare system relies on vulnerable fragmented, centralized databases
- limited patient control

MECHANISM

- Smart contracts: AUTOMATED AUDITORS
- conditionally granting data access and verifying treatments to eliminate systemic fraud.

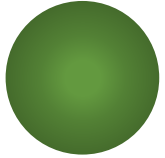
SOLUTION

- Blockchain transitions healthcare to a decentralized, encrypted network

ECONOMIC REALITY

- shifts the payoff matrix to mathematically
- secures digital data

9. Key Takeaways



Vulnerability

Centralized healthcare systems create single points of failure that lead to massive data breaches and fragmented patient care



Patient Sovereignty:

- Blockchain decoupling patient's health records from hospital silos and giving them direct cryptographic control.
- Smart contracts mathematically neutralize financial fraud (upcoding, trial manipulation)



The Ultimate Limit:

Blockchain guarantees digital accountability and data integrity BUT cannot verify the physical quality of care provided

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